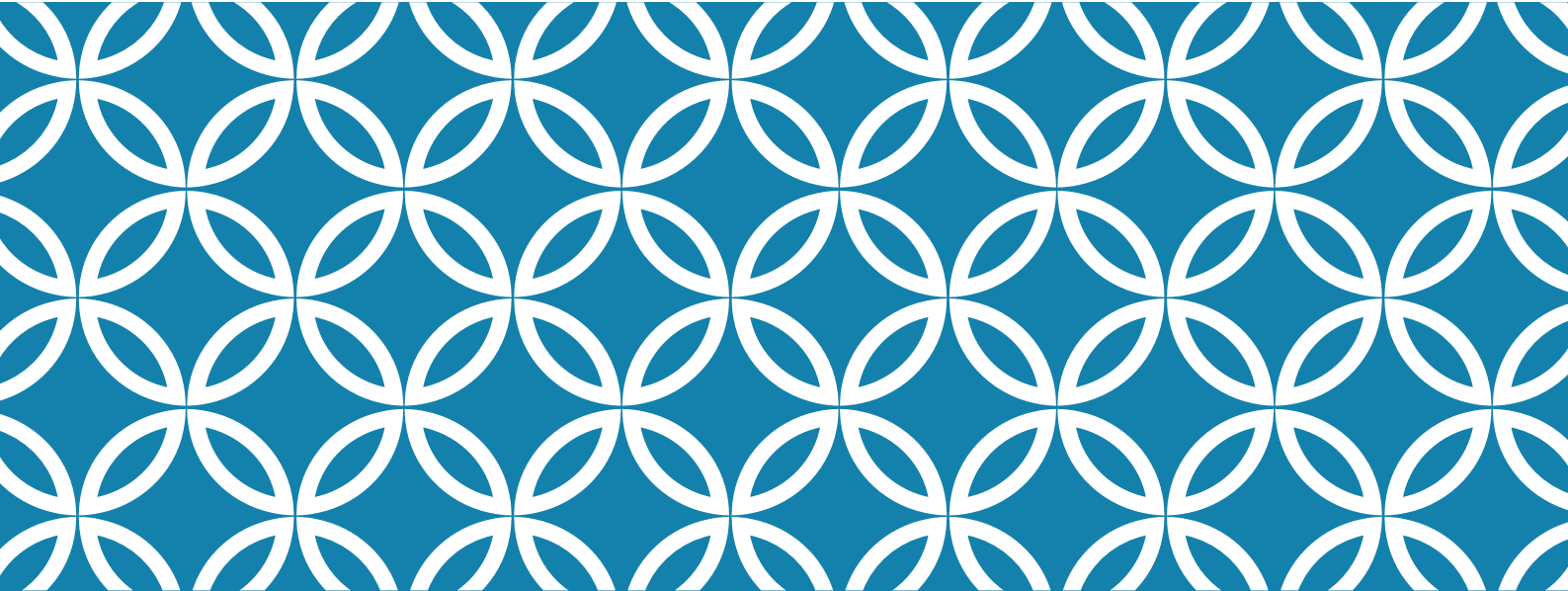




5.3

Higher-Order Taylor Methods

Formula

Taylor method of order n

$$w_0 = \alpha,$$

$$w_{i+1} = w_i + hT^{(n)}(t_i, w_i), \quad \text{for each } i = 0, 1, \dots, N-1, \quad (5.17)$$

where

$$T^{(n)}(t_i, w_i) = f(t_i, w_i) + \frac{h}{2}f'(t_i, w_i) + \dots + \frac{h^{n-1}}{n!}f^{(n-1)}(t_i, w_i).$$

Euler's method is Taylor's method of order one.

EXERCISE SET 5.3

1. Use Taylor's method of order two to approximate the solutions for each of the following initial-value problems.
 - a. $y' = te^{3t} - 2y$, $0 \leq t \leq 1$, $y(0) = 0$, with $h = 0.5$
 - b. $y' = 1 + (t - y)^2$, $2 \leq t \leq 3$, $y(2) = 1$, with $h = 0.5$
 - c. $y' = 1 + y/t$, $1 \leq t \leq 2$, $y(1) = 2$, with $h = 0.25$
 - d. $y' = \cos 2t + \sin 3t$, $0 \leq t \leq 1$, $y(0) = 1$, with $h = 0.25$
3. Repeat Exercise 1 using Taylor's method of order four.

$$1) a) f(t, y) = t e^{3t} - 2y$$

For order 2: $T^2(t_i, w_i) = f(t_i, w_i) + \frac{h}{2} f'(t_i, w_i)$

$$f'(t_i, w_i) = \frac{d}{dt} [f(t, w_i)]$$

$$(3t)' = 3$$

$$\frac{d}{dt} [f(t, t)] = \frac{d}{dt} (t e^{3t} - 2t) = 3e^{3t}$$

$$= 1 e^{3t} + t \cdot 3e^{3t} - 2$$

$$= e^{3t} + 3t e^{3t} - 2(t e^{3t} - 2t)$$

$$= e^{3t} + 3t e^{3t} - 2t e^{3t} + 4t$$

$$= e^{3t} + t e^{3t} + 4t$$

$$f(t, w) = t e^{3t} - 2w$$

$$f(t_i, w_i) = t_i e^{3t_i} - 2w_i$$

$$f'(t, w) = e^{3t} + t e^{3t} + 4w$$

$$f'(t_i, w_i) = e^{3t_i} + t_i e^{3t_i} + 4w_i$$

$$T^2(t_i, w_i) = f(t_i, w_i) + \frac{h}{2} f'(t_i, w_i)$$

$$= t_i e^{3t_i} - 2w_i + \frac{0.5}{2} [e^{3t_i} + t_i e^{3t_i} + 4w_i]$$

$$u v = u' v + v' u$$

t_i	w_i
$t_0 = 0$	$w_0 = 0$
$t_1 = 0.5$	$w_1 = 0.125$
$t_2 = 1$	$w_2 = 2.023238$

$$w_{i+1} = w_i + h T^2(t_i, w_i)$$

$$= w_i + h [t_i e^{3t_i} - 2w_i + \frac{h}{2} [e^{3t_i} + t_i e^{3t_i} + 4w_i]]$$

$$w_1 = w_0 + 0.5 [t_0 e^{3t_0} - 2w_0 + \frac{0.5}{2} [e^{3t_0} + t_0 e^{3t_0} + 4w_0]]$$

$$= 0 + 0.5 [0 e^{3(0)} - 2(0) + \frac{0.5}{2} [e^{3(0)} + 0 e^{3(0)} + 4(0)]] = 0.125$$

$$w_2 = w_1 + 0.5 [t_1 e^{3t_1} - 2w_1 + \frac{0.5}{2} [e^{3t_1} + t_1 e^{3t_1} + 4w_1]]$$

$$= 0.125 + 0.5 [0.5 e^{3(0.5)} - 2(0.125) + \frac{0.5}{2} [e^{3(0.5)} + 0.5 e^{3(0.5)} + 4(0.125)]] = 2.023238$$

b. $y' = 1 + (t - y)^2$, $2 \leq t \leq 3$, $y(2) = 1$, with $h = 0.5$

$$g(t, y) = 1 + (t - y)^2 \quad w_0 = 1$$

$$\tau^2(t_i, w_i) = g(t_i, w_i) + \frac{h}{2} g'(t_i, w_i)$$

$$g'(t_i, w_i) = \frac{d}{dt} [g(t_i, w_i)]$$

$$\frac{d}{dt} [g(t, y)] = 2(t - y)(1 - y') = 2(t - y)(1 - [1 + (t - y)^2]) = 2(t - y)(- (t - y)^2) = -2(t - y)^3 = -2(t - y)^3$$

$$g'(t_i, w_i) = -2(t_i - w_i)^3$$

$$w_{i+1} = w_i + h \tau^2(t_i, w_i) \\ = w_i + h \left[1 + (t_i - w_i)^2 + \frac{h}{2} [-2(t_i - w_i)^3] \right]$$

$$w_1 = w_0 + h \left[1 + (t_0 - w_0)^2 + \frac{h}{2} [-2(t_0 - w_0)^3] \right] \\ = 1 + 0.5 \left[1 + (2 - 1)^2 + \frac{0.5}{2} [-2(2 - 1)^3] \right] \\ = 1.75$$

$$w_2 = w_1 + h \left[1 + (t_1 - w_1)^2 + \frac{h}{2} [-2(t_1 - w_1)^3] \right] \\ = 1.75 + 0.5 \left[1 + (2.5 - 1.75)^2 + \frac{0.5}{2} [-2(2.5 - 1.75)^3] \right] \\ = 2.425781$$

t_i	w_i
$t_0 = 2$	$w_0 = 1$
$t_1 = 2.5$	$w_1 = 1.75$
$t_2 = 3$	$w_2 =$

3d Taylor's method of order 4
 $0 \leq t \leq 1$, $y(0)=1$, $h=0.25$

$$y' = \cos 2t + \sin 3t$$

$$w_0 = y(0) = 1$$

$$f(t, y) = \cos 2t + \sin 3t \Rightarrow f(t_i, w_i) = \cos 2t_i + \sin 3t_i$$

$$f'(t, y) = \frac{d}{dt} (\cos 2t + \sin 3t) = -2 \sin 2t + 3 \cos 3t$$

$$f''(t, y) = \frac{d}{dt} (-2 \sin 2t + 3 \cos 3t) = -4 \cos 2t - 9 \sin 3t$$

$$f'''(t, y) = \frac{d}{dt} (-4 \cos 2t - 9 \sin 3t) = 8 \sin 2t - 27 \cos 3t$$

$$\boxed{n=4} \quad T^4(t_i, w_i) = f(t_i, w_i) + \frac{h}{2} f'(t_i, w_i) + \frac{h^2}{3!} f''(t_i, w_i) + \frac{h^3}{4!} f'''(t_i, w_i)$$

$$= \cos 2t_i + \sin 3t_i + \frac{h}{2} [-2 \sin 2t_i + 3 \cos 3t_i] + \frac{h^2}{6} [-4 \cos 2t_i - 9 \sin 3t_i] + \frac{h^3}{24} [8 \sin 2t_i - 27 \cos 3t_i]$$

$$w_{i+1} = w_i + h T^4(t_i, w_i)$$

$$w_{i+1} = w_i + 0.25 \left[\cos 2t_i + \sin 3t_i + \frac{0.25}{2} (-2 \sin 2t_i + 3 \cos 3t_i) + \frac{0.25^2}{6} (-4 \cos 2t_i - 9 \sin 3t_i) + \frac{0.25^3}{24} (8 \sin 2t_i - 27 \cos 3t_i) \right]$$

$$w_1 = w_0 + 0.25 \left[\cos 2t_0 + \sin 3t_0 + \frac{0.25}{2} (-2 \sin 2t_0 + 3 \cos 3t_0) + \frac{0.25^2}{6} (-4 \cos 2t_0 - 9 \sin 3t_0) + \frac{0.25^3}{24} (8 \sin 2t_0 - 27 \cos 3t_0) \right]$$

$$w_1 = 1 + 0.25 \left[\cos 2(0) + \sin 3(0) + \frac{0.25}{2} (3 \cos 3(0)) + \frac{0.25^2}{6} (-4 \cos 2(0)) + \frac{0.25^3}{24} (-27 \cos 3(0)) \right] = 1.32894$$

$$w_2 = 1.32894 + 0.25 \left[\cos 2(0.25) + \sin 3(0.25) + \frac{0.25}{2} (-2 \sin 2(0.25) + 3 \cos 3(0.25)) + \frac{0.25^2}{6} (-4 \cos 2(0.25) - 9 \sin 3(0.25)) + \frac{0.25^3}{24} (8 \sin 2(0.25) - 27 \cos 3(0.25)) \right]$$

$$= 1.72966$$

t_i	w_i
$t_0 = 0$	$w_0 = 1$
$t_1 = 0.25$	$w_1 = 1.32894$
$t_2 = 0.5$	$w_2 = 1.72966$
$t_3 = 0.75$	w_3
$t_4 = 1$	w_4

$$\omega_{i+1} = \omega_i + h \tau^4(t_i, \omega_i)$$

$$\omega_{i+1} = \omega_i + 0.25 \left[\cos 2t_i + \sin 3t_i + \frac{0.25}{2} (-2 \sin 2t_i + 3 \cos 3t_i) + \frac{0.25^2}{6} (-4 \cos 2t_i - 9 \sin 3t_i) + \frac{(0.25)^3}{24} (8 \sin 2t_i - 27 \cos 3t_i) \right]$$

$$\omega_1 = \omega_0 + 0.25 \left[\cos 2t_0 + \sin 3t_0 + \frac{0.25}{2} (-2 \sin 2t_0 + 3 \cos 3t_0) + \frac{0.25^2}{6} (-4 \cos 2t_0 - 9 \sin 3t_0) + \frac{0.25^3}{24} (8 \sin 2t_0 - 27 \cos 3t_0) \right]$$

$$\omega_1 = 1.72966 + 0.25 \left[\cos 2(0.5) + \sin 3(0.5) + \frac{0.25}{2} (-2 \sin 2(0.5) + 3 \cos 3(0.5)) + \frac{0.25^2}{6} (-4 \cos 2(0.5) - 9 \sin 3(0.5)) + \frac{0.25^3}{24} (8 \sin 2(0.5) - 27 \cos 3(0.5)) \right] = 2.03993$$

$$\omega_4 = 2.03993 + 0.25 \left[\cos 2(0.75) + \sin 3(0.75) + \frac{0.25}{2} (-2 \sin 2(0.75) + 3 \cos 3(0.75)) + \frac{0.25^2}{6} (-4 \cos 2(0.75) - 9 \sin 3(0.75)) + \frac{0.25^3}{24} (8 \sin 2(0.75) - 27 \cos 3(0.75)) \right] = 2.11599$$

t_i	ω_i
$t_0 = 0$	$\omega_0 = 1$
$t_1 = 0.25$	$\omega_1 = 1.72966$
$t_2 = 0.5$	$\omega_2 = 2.03993$
$t_3 = 0.75$	$\omega_3 = 2.11599$
$t_4 = 1$	$\omega_4 = 2.11599$

$$(3a) \quad y' = te^{3t} - 2y, \quad 0 \leq t \leq 1, \quad t=0.5, \quad y(0)=0$$

$$f(t, y) = te^{3t} - 2y$$

$$f'(t, y) = 1 \cdot e^{3t} + t(3e^{3t}) - 2y' = e^{3t} + 3te^{3t} - 2(te^{3t} - 2y) \\ = e^{3t} + 3te^{3t} - 2te^{3t} + 4y = \boxed{e^{3t} + te^{3t} + 4y}$$

$$f''(t, y) = 3e^{3t} + 1 \cdot e^{3t} + t(3e^{3t}) + 4y' = 4e^{3t} + 3te^{3t} + 4(te^{3t} - 2y) \\ = 4e^{3t} + 3te^{3t} + 4te^{3t} - 8y = \boxed{4e^{3t} + 7te^{3t} - 8y}$$

$$f'''(t, y) = 12e^{3t} + 7(1 \cdot e^{3t} + t \cdot 3e^{3t}) - 8y' = 12e^{3t} + 7e^{3t} + 21te^{3t} - 8y' \\ = 19e^{3t} + 21te^{3t} - 8(te^{3t} - 2y) \\ = 19e^{3t} + 21te^{3t} - 8te^{3t} + 16y = \boxed{19e^{3t} + 13te^{3t} + 16y}$$

$$w_{i+1} = w_i + h \left[f(t_i, w_i) + \frac{h}{2} f'(t_i, w_i) + \frac{h^2}{3!} f''(t_i, w_i) + \frac{h^3}{4!} f'''(t_i, w_i) \right]$$

$$w_2 = w_1 + 0.5 \left[t_1 e^{3t_1} - 2w_1 + \frac{0.5}{2} (e^{3t_1} + t_1 e^{3t_1} + 4w_1) \right. \\ \left. + \frac{0.5^2}{6} (4e^{3t_1} + 7t_1 e^{3t_1} - 8w_1) \right. \\ \left. + \frac{0.5^3}{24} (19e^{3t_1} + 13t_1 e^{3t_1} + 16w_1) \right]$$

$$= 0.2578125 + 0.5 \left[0.5e^{3(0.5)} - 2(0.2578125) + \frac{0.5}{2} (e^{3(0.5)} + 0.5e^{3(0.5)} + 4(0.2578125)) \right. \\ \left. + \frac{0.5^2}{6} (4e^{3(0.5)} + 7(0.5)e^{3(0.5)} - 8(0.2578125)) \right. \\ \left. + \frac{0.5^3}{24} (19e^{3(0.5)} + 13(0.5)e^{3(0.5)} + 16(0.2578125)) \right] \\ = 3.055295$$